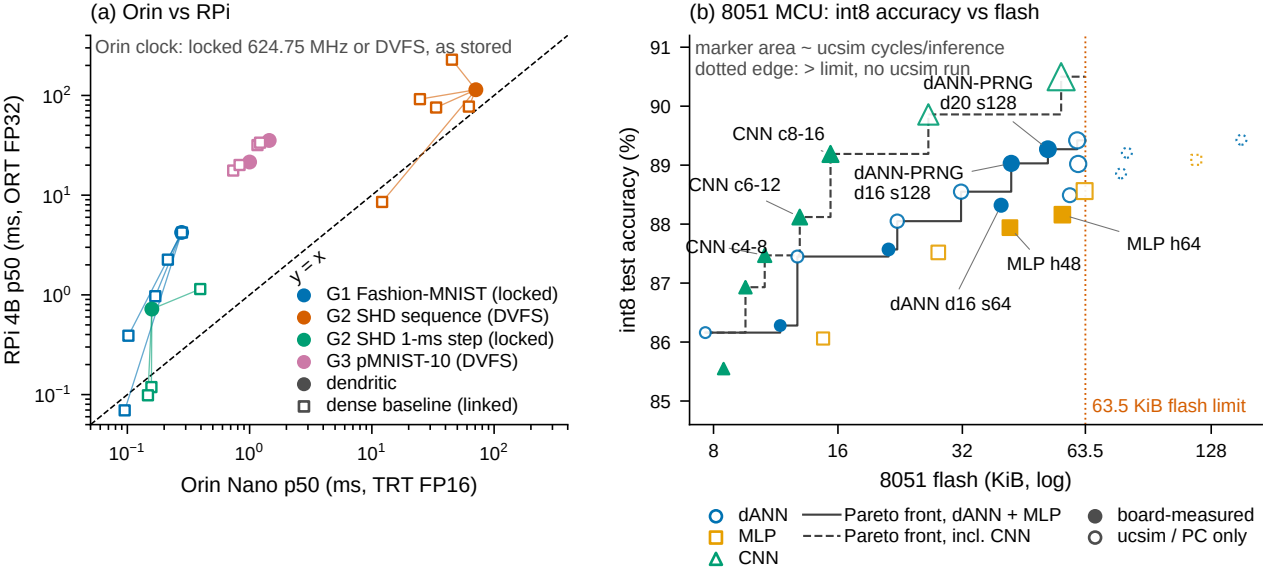
(a) Orin vs RPi
Orin clock: locked 624.75 MHz or DVFS, as stored
RPi 4B p50 (ms, ORT FP32)
Orin Nano p50 (ms, TRT FP16)
y = x
G1 Fashion-MNIST (locked)
G2 SHD sequence (DVFS)
G2 SHD 1-ms step (locked)
G3 pMNIST-10 (DVFS)
dendritic
dense baseline (linked)
(b) 8051 MCU: int8 accuracy vs flash
marker area ~ ucsim cycles/inference
dotted edge: > limit, no ucsim run
int8 test accuracy (%)
8051 flash (KiB, log)
dANN-PRNG d20 s128
CNN c8-16
dANN-PRNG d16 s128
CNN c6-12
CNN c4-8
MLP h64
MLP h48
dANN d16 s64
63.5 KiB flash limit
dANN
MLP
CNN
Pareto front, dANN + MLP
Pareto front, incl. CNN
board-measured
ucsim / PC only